BEACON INNOVATION HUB

DATA DEPARTMENT FOUNDATIONAL COMPETENCE CHALLENGE

Edu-ca-te Learner Data & Resource Allocation Case Study

Client	Mr P.G. Marapira
Pathways	Data Analyst | Data Scientist | Data Engineer
Level	Foundational / Junior Data Competence
Core Decision	Which subject should receive additional tutoring resources next term?

1. Business Scenario

Edu-ca-te is a tutoring platform supporting Grade 10-12 learners across several school subjects. As the platform grows, management must decide how limited tutoring resources should be distributed for the next academic term. Additional tutoring hours require money, tutor availability and operational capacity, so the decision cannot be based on intuition or a single measure such as learner count.

Edu-ca-te has collected learner registration information, engagement activity, assessment results, tutoring capacity information and learner enquiries. These records originate from different operational processes and were not designed as one clean analytical system. As a result, the supplied workbook contains realistic data-quality problems that may affect calculations, joins and business conclusions.

Beacon Innovation Hub has been asked to investigate whether the data can support a defensible resource-allocation decision. The candidate acts as a junior data professional assigned to turn the raw operational records into reliable evidence. Management wants to know where learner need is greatest, whether learners appear to be benefiting from the service, where demand is increasing, and where current tutoring capacity may be insufficient.

The final recommendation must therefore be derived from multiple pieces of evidence. A subject with many learners may already have sufficient tutoring capacity. A subject with strong enquiry demand may have weak conversion. A subject with low average performance may not necessarily have the largest waiting list. Candidates are expected to reconcile these signals rather than selecting whichever subject happens to rank first on one metric.

2. Supplied Data

learners: Learner master information including learner identifier, gender, grade and registration date.

engagement: Subject-level learner participation including attendance, videos watched and watch time.

assessments: Baseline and latest learner assessment marks by subject.

support_capacity: Current tutoring capacity by subject, including active tutors, weekly tutoring hours, waiting-list demand and estimated cost of additional tutoring time.

enquiries: Prospective learner enquiries by date, grade and subject, together with an indication of whether the enquiry converted to registration.

The workbook is **intentionally imperfect**. Candidates should expect missing values, duplicated or conflicting records, inconsistent identifiers and category labels, mixed formats, suspicious numerical values, invalid dates, unmatched identifiers and other quality problems. The location and significance of these problems are part of the assessment and are not provided as a checklist.

3. Management Decision

Edu-ca-te can fund additional tutoring resources, but management wants the investment directed toward the subject where the evidence indicates the strongest operational need. The candidate must determine which subject should receive priority and defend the recommendation.

Primary business question: Which subject should receive additional tutoring resources next term, and why?

A credible answer should consider several relevant signals where the data permits, including current learner participation, attendance and engagement, assessment performance and mark change, enquiry demand, enquiry conversion, waiting-list pressure, existing tutor capacity, available weekly tutoring hours and the estimated cost of expanding tutoring.

4. Candidate Investigation

- **Inspect the workbook** before changing any data. Explain what each dataset represents, its likely unit of observation and how the datasets relate.
- Perform a **data-quality audit**. Identify and quantify important quality problems and explain how they could distort analysis or the final business recommendation.

- **Clean and standardize** the data using defensible rules. Maintain a cleaning log that records the problem, action and justification. Do not silently delete inconvenient observations or fabricate missing information.
- **Validate the cleaned data**. Define reasonable rules for learner IDs, grades, subjects, dates, assessment marks, attendance, video activity, watch time, capacity and cost fields.
- Determine the **correct keys** and **join strategy**. Demonstrate that joins have not unintentionally multiplied records or removed important observations.
- Measure learner participation and engagement by subject and grade. Identify important differences rather than reporting totals without interpretation.
- Investigate whether attendance, videos watched or watch time appear related to assessment change. Clearly **distinguish association from causation**.
- Analyse **enquiry demand** and conversion by subject. Determine whether prospective demand tells a different story from current registered participation.
- Evaluate **tutoring capacity**. Compare learner demand and waiting-list pressure with active tutors and weekly tutor hours. Consider the cost of adding capacity.
- Develop a **small set of decision-relevant KPIs** that allow management to compare subjects. Explain why each KPI belongs in the decision and avoid creating an unexplained arbitrary score.
- Recommend which subject should receive additional tutoring resources. Support the recommendation with cleaned-data evidence and explain important trade-offs, assumptions and limitations.
- Explain **what additional data Edu-ca-te should collect** before making larger or longer-term resource-allocation decisions.

5. Pathway Emphasis

Pathway	Primary Competence	Expected Evidence
Data Analyst	Clean, summarize, visualize and communicate business information.	Reliable KPIs, comparisons, charts and a clear resource-allocation recommendation.
Data Scientist	Investigate relationships and reason carefully from evidence.	Derived variables, pattern investigation, uncertainty, alternative explanations and cautious conclusions.
Data Engineer	Structure, validate and move data reliably.	Schema design, keys, validation rules, reproducible transformations and a proposed scalable data flow.

6. Submission Requirements (DELIVERABLES)

- **Cleaned datasets** and/or a reproducible cleaning script or notebook.
- A **data-quality audit** and cleaning log.
- **Validation evidence** and join-integrity checks.
- Analysis used to answer the pathway-relevant questions.
- At least one clear **management-facing visualization**.
- A **concise business recommendation** supported by data.
- A limitations section separating evidence from assumptions.
- For Data Engineer candidates, a **basic relational schema** and proposed raw-to-validated data flow.
- An organized repository or submission that allows a BIH reviewer to reproduce the work.

7. Assessment Principle

This assessment measures competence in working with imperfect real-world data. Candidates are not rewarded merely for producing attractive dashboards or correct-looking numbers. Credit should be based on whether the candidate can identify data problems, make defensible cleaning decisions, validate the result, preserve join integrity, select appropriate analytical methods, interpret evidence in business context and reproduce the work.

During technical defence, a BIH reviewer may select any cleaning rule, join, KPI, calculation, visualization or recommendation and ask the candidate to explain exactly how it was produced and why the decision was reasonable. A candidate who cannot explain or reproduce a result should not receive full competence credit for that result.

8. Important Constraint

There is no requirement for advanced machine learning. The challenge is designed to test foundational data competence, analytical judgement and business reasoning. Candidates should use the simplest appropriate method that produces reliable and explainable evidence.

ENDNOTES

We wish you all the best candidates as you embark on this wonderful journey. SUBMISSION date is Wednesday 26 August 2026. The assessment is to be done individually, and all files must be submitted on a GitHub repository.

GOODLUCK!!!

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